

Experiment0

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1 Part 1: DC Analysis

1.1 a)

Method 1: By Ohm's Law, given voltage and current, we can calculate the resistance. Here I set the voltage source to be 18V. The current is 1.2mA as shown.

$$Req = V_s/I = 18V/1.2mA = 15k\Omega$$

Method 2: Instead of using a voltage source and measuring the current, we can also obtain the resistance by measuring the voltage, given current source. Here, I set the current source to be 1A.

$$Req = V/I_s = 15kV/1A = 15k\Omega$$

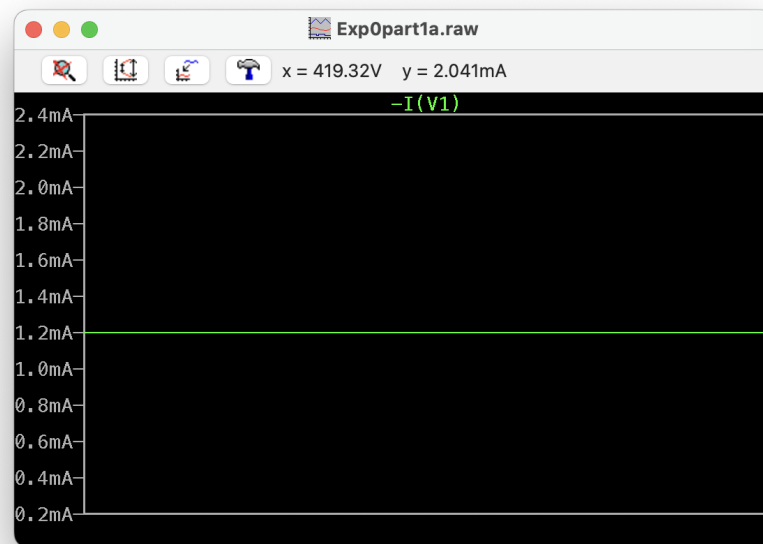
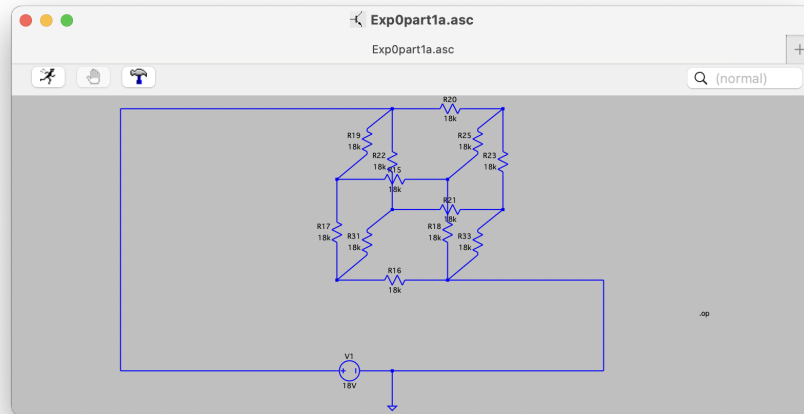


Figure 1: $R_{eq} = V/I = 18V/1.2mA = 15\text{ k}\Omega$

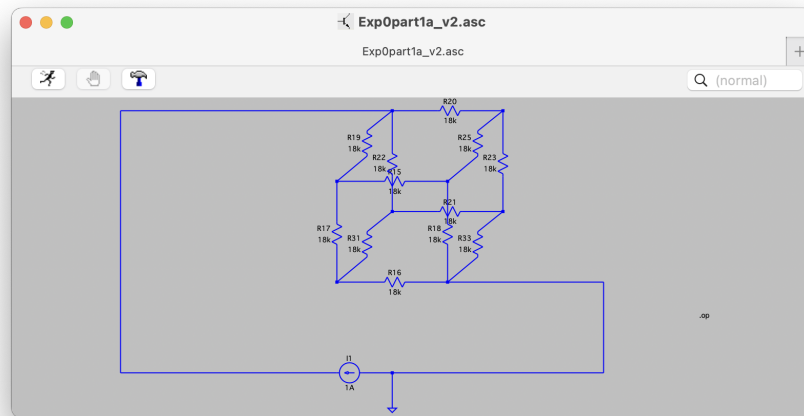
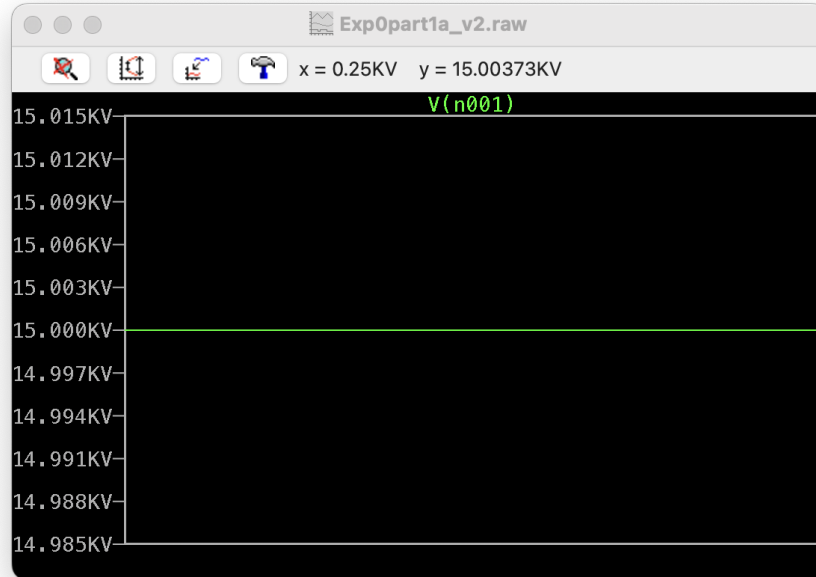


Figure 2: $R_{eq} = V/I_s = 15\text{kV}/1\text{A} = 15\text{ k } \Omega$

1.2 b)

We can see a concave down power plot in both cases. At critical point, where $R_{load} = 100 \Omega$, maximum power is delivered.

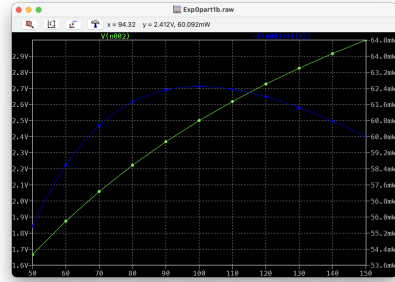


Figure 3: 50-150 Ω , $R_{load} = 100 \Omega$

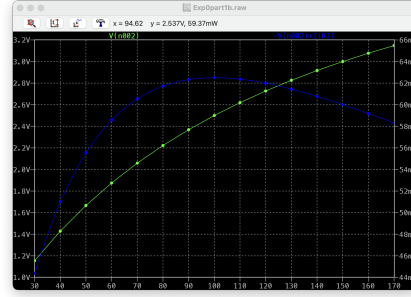


Figure 4: 30-170 Ω , $R_{load} = 100 \Omega$

1.3 c)

The diode needs 0.7 V forward bias to conduct the current. Before 0.7 V, the V_{out} is just the same as V_{in} as no current flow through the circuit (open circuit). V_{out} is the voltage across the diode. Once the voltage crosses the 0.7, the diode began to conduct the current and therefore the V_{out} is decreasing. The diode behaves "more like a short circuit". Therefore, the voltage across it decreases.

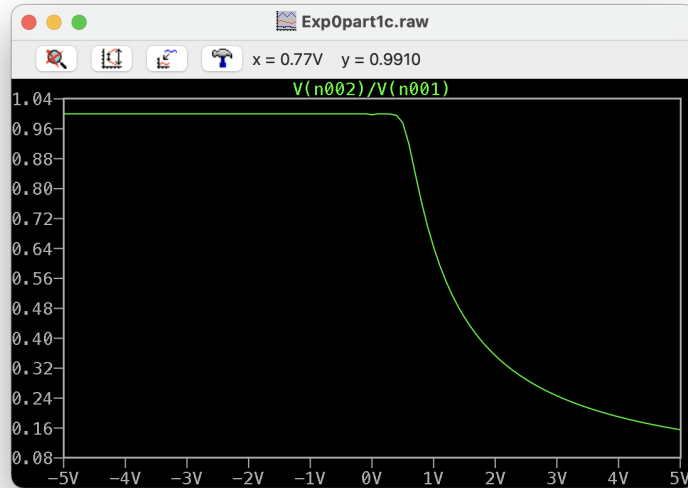


Figure 5: Over the range: $-5 \text{ volts} < V_{in} < 5 \text{ volts}$, with $0.1V$ increment

2 Part 2: Transient

2.1 a)

The distortion is changing because the signal's amplitude is changing. In the first analysis where the voltage peak is 1 V, 12.76% distortion happens. The second analysis where the voltage peak is 2V has a 29.20% distortion. The circuit allows a certain amplitude of signal to pass through. In this case, partial signal under 0.7V is past. More distortion will happen if we continue to increase the amplitude while keeping the circuit design the same.

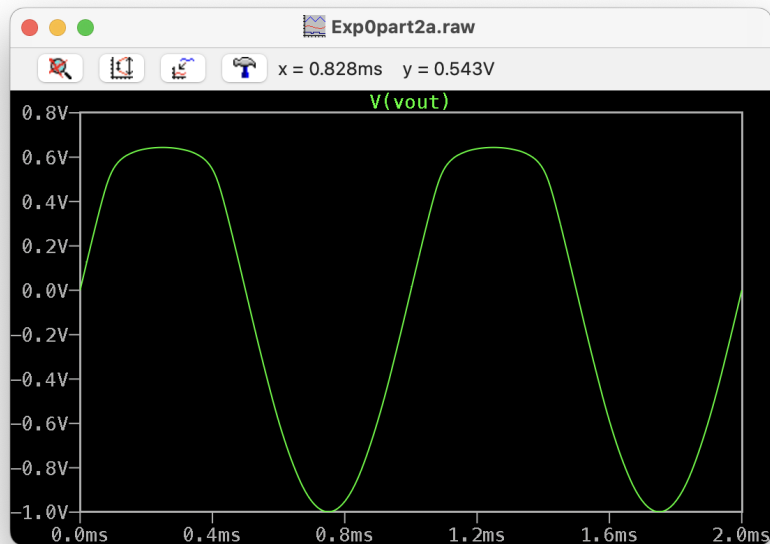


Figure 6: V_{in} with a sinusoidal source of frequency 1000Hz and peak value 1V.
Harmonic distortion 1: 12.76%

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Ltpice 17.2.4 for Mac05
Circuit: /Users/lab/Desktop/Columbia University/Courses/ELN E3201 Circuit Analysis/Lab/
Ltpicefile/Exp0part2a.as
Start Time: Mon Sep 22 15:32:52 2025
solver = Normal
Maximum thread count: 16
from = 27
temp = 27
method = modified trapezoidal
CompressMaxPoints = 3024
JP point found by inspection.
N-Periods:
Fourier components of V(vout)
DC component = 0.072522

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Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [deg]	Normalized Phase [deg]
1	1.000e+3	8.633e-1	1.000e+0	86.81°	0.00°
2	2.000e+3	9.565e-2	1.102e-1	8.18°	-89.98°
3	3.000e+3	5.269e-2	6.045e-2	89.75°	-8.50°
4	4.000e+3	1.638e-2	1.887e-2	-178.58°	-205.99°
5	5.000e+3	3.354e-3	3.886e-3	92.38°	3.38°
6	6.000e+3	8.331e-3	9.546e-3	178.00°	88.89°
7	7.000e+3	4.229e-3	4.777e-3	-98.01°	-188.42°
8	8.000e+3	1.777e-4	1.999e-4	111.94°	43.93°
9	9.000e+3	2.748e-3	3.156e-3	-85.15°	-175.15°
10	1.000e+4	1.472e-3	1.680e-3	-6.80°	-26.80°
11	1.100e+4	8.481e-5	9.674e-5	-81.91°	-171.91°
12	1.200e+4	8.653e-4	9.774e-4	-18.48°	-108.49°
13	1.300e+4	8.966e-4	1.033e-3	181.74°	11.74°
14	1.400e+4	3.753e-5	4.323e-5	56.13°	-33.88°
15	1.500e+4	4.292e-4	4.943e-4	94.86°	4.85°
16	1.600e+4	2.588e-4	2.979e-4	161.13°	71.13°
17	1.700e+4	5.581e-6	6.335e-6	-61.66°	-151.67°
18	1.800e+4	1.571e-4	1.800e-4	178.68°	88.68°
19	1.900e+4	1.391e-4	1.581e-4	-187.82°	-197.82°
20	2.000e+4	2.419e-5	2.786e-5	78.38°	-19.62°

Partial Harmonic Distortion: 12.764894%

Total Harmonic Distortion: 12.764164%

Total elapsed time: 0.012 seconds.

Figure 7: V_{in} with a sinusoidal source of frequency 1000Hz and peak value 1V.
Harmonic distortion 1: 12.76%

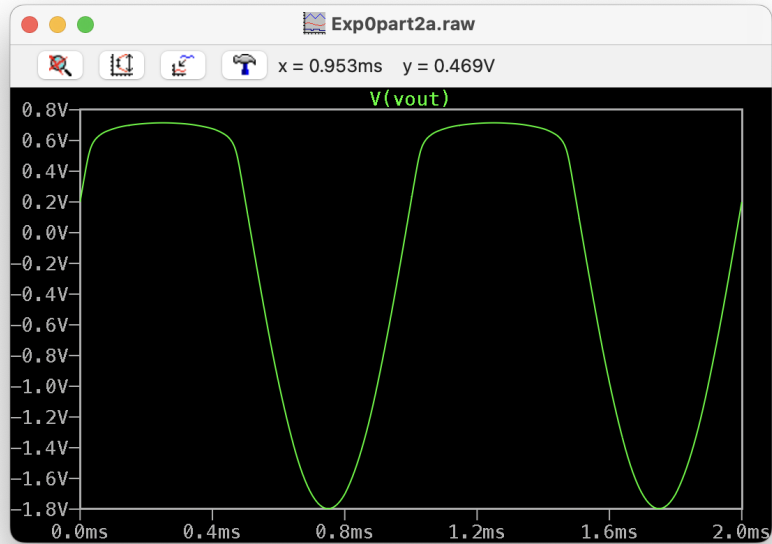


Figure 8: V_{in} with a sinusoidal source of frequency 1000Hz and peak value 2V.
Harmonic distortion 1: 29.20%

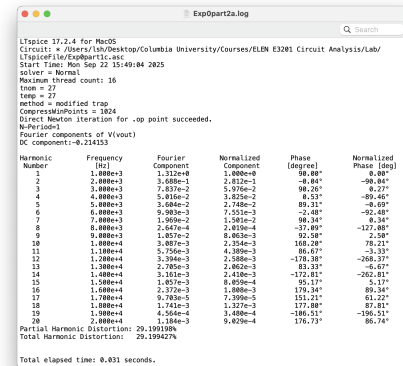


Figure 9: V_{in} with a sinusoidal source of frequency 1000Hz and peak value 2V.
Harmonic distortion 1: 29.20%

2.2 b)

When the output voltage has completed 63% of its total change, $V_{out} = 5.92\mu V$. By placing the cursor where the V_{out} is $5.92\mu V$, we can observe that the time constant is 0.48 ms. This result matches our theoretical value $\tau = RC = 100\Omega \times 5\mu F = 500$.

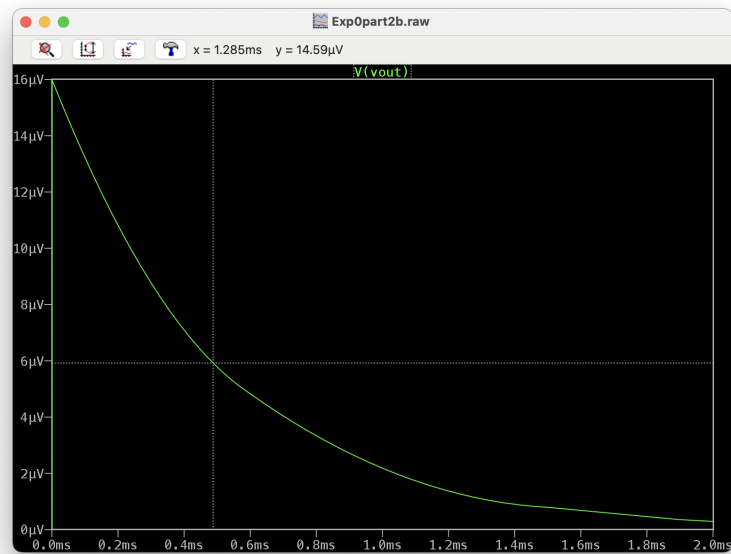


Figure 10: Cursor position is $5.92\mu V$ at 0.48ms. $\tau = 0.48$

3 Part 3: AC Analysis

This is a low pass filter. The cutoff frequency is at -3dB. In the low frequency range, most signal can pass the filter. Beyond the 0.1Hz, the gain of the filter decrease in a slope of -20dB/decade. That means, higher frequency are filtered out. We can also see that the phase plot has a range of 0 to -180 degrees.

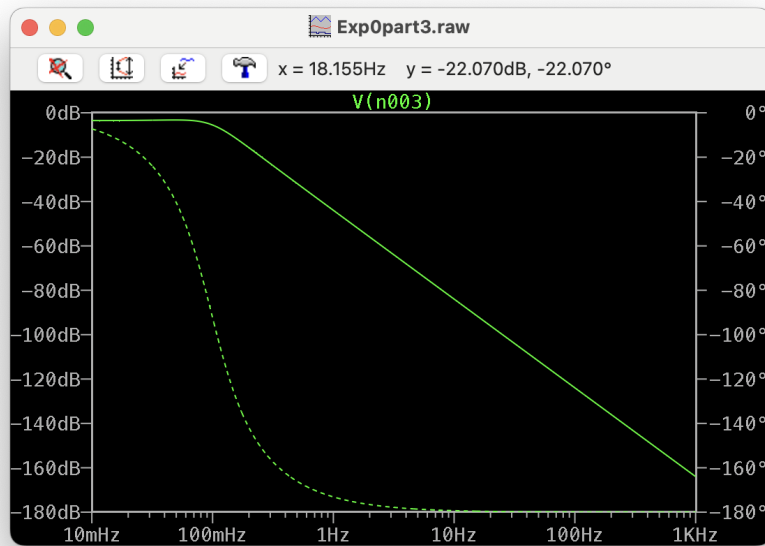


Figure 11: Frequency response over the range of 0.01 Hz to 1 kHz